

MovementLogger — Sensor & Module Catalog

10 Teile · 2026-05-21 11:16 UTC · BOM aus movement_logger_firmware · via crawl2pump

10 neu (7 Tage)

0 aktualisiert

· 99 touched · 0 price changes

Dev Boards 4

STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C)

OSS firmware

USB-C pluggable

☒ USB-C

☐ WiFi

☒ Bluetooth

☐ GPS

☒ Motion sensors

☒ SD-card

L×B×H 6.3 × 4 × 2 cm · 94 g

MCU: STM32U585 · Cortex-M33 @160 MHz · 2 MB flash / 786 KB SRAM · TrustZone · ultra-low-power

Datasheet: https://www.st.com/resource/en/user_manual/um3133-sensortilebox-pro-stmicroelectronics.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C) · STEVAL-MKBOXPRO @ DigiKey

CHF 94.56

STMicroelectronics @ DigiKey · in stock

Primary target board. USB-C for charge + STM32 DFU flashing. · USB-C · firmware: open-source

<https://www.digikey.com/en/products/detail/stmicroelectronics/STEVAL-MKBOXPRO/19721662>

GPS / GNSS 18

AliExpress FC 14-pin Cable — 1.27 mm IDC → DuPont female, 30 cm (2X7P variant)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 30 × 0.5 × 0.3 cm · 7 g



neu AliExpress FC 14-pin Cable — 1.27 mm IDC → DuPont female, 30 cm (2X7P variant) · Vino Electronic Connect. (AliExpress) @ Vino Electronic Connect. (AliExpress) (vendor page)

Vino Electronic Connect. (AliExpress) (manufacturer store)

****PIN CHOICE IS CRITICAL — wrong pins can brick SWD debug or destroy GPS**** Use ONLY 3 of the 14 wires: • ****pin 13**** (UART4_RX on STEVAL) → GPS ****TX**** • ****pin 14**** (UART4_TX on STEVAL) → GPS ****RX**** • ****pin 7 or 11**** (GND) → GPS ****GND**** ALL OTHER PINS carry SWD/JTAG signals — ****do NOT touch**** (bridging any of them can permanently lock out debug access or damage the GPS). Identify pin 1 by the red/blue stripe on the ribbon (standard IDC marking — pin 1 is on the same side as the marked wire). The header is 2-row × 7-col: odd pins (1,3,5,7,9,11,13) on one row, even pins (2,4,6,8,10,12,14) on the other; pin 13/14 are the LAST pair at the far end of the connector. If unsure, multimeter continuity test: STEVAL off, probe JP2 pin 1 silkscreen marker against each wire end. GPS VCC (3.3 V) comes from JP4, NOT JP2 — use the Samtec SSW-107-01-G-S header card below + one F/F DuPont from the...

<https://de.aliexpress.com/item/1005011846708876.html>

GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 4 × 4 × 1.3 cm · 80 g

Datasheet: <https://www.sparkfun.com/products/14986>



neu **GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) · GPS-14986 @ DigiKey**

SparkFun @ DigiKey · out of stock

Active GPS antenna (~28 dB LNA, 3 m RG-174 coax, SMA male). Magnetic base — sits on the board's non-metal top or the case lid for best sky view. Needs a U.FL → SMA pigtail to mate with the MAX-M10S breakout's U.FL connector — see next part. · SMA / U.FL · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/14986/9682235>

CHF 12.85

Interface Cable SMA → U.FL, 100 mm (pigtail) **OSS firmware** **SMA / U.FL**

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: <https://www.sparkfun.com/products/9145>



neu **Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com**

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL · firmware: open-source

<https://www.sparkfun.com/products/9145>

USD 5.75

Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (JP4 power tap) **OSS firmware**

SMA / U.FL

☐ USB-C

☐ WiFi

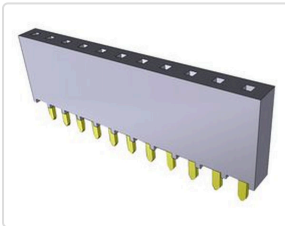
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 1.8 × 0.85 × 0.25 cm · 0.50 g



neu **Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (JP4 power tap) · SSW-107-01-G-S @ Mouser**

Samtec @ Mouser · in stock

1×7 gold-plated through-hole socket strip on 2.54 mm (0.1") pitch. Slips onto JP4, the STEVAL's 7-pin power header — no soldering, friction-fit. Before plugging anything in: **set the JP4 domain switch to 3 V** (not 1.8 V) and meter the 3 V pin under power; must read 3.25–3.40 V. Then DuPont-jumper the 3 V pin to the SparkFun MAX-M10S breakout's VCC pin. The strip's other 6 pins stay accessible for GND tapping or for future signal-monitoring. · SMA / U.FL · firmware: open-source

<https://www.mouser.ch/ProductDetail/Samtec/SSW-107-01-G-S?qs=rU5fayqh%252BE2WDJ3AF5mWdA%3D%3D>

CHF 1.34

SparkFun GPS Breakout - MAX-M10S (Qwiic) **OSS firmware** **Qwiic / I²C**

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

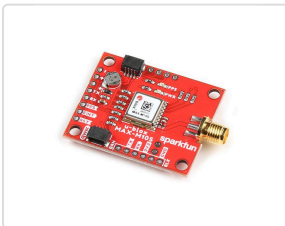
☐ Motion sensors

☐ SD-card

L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu **SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey**

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. Has an onboard chip antenna that works through the polycarbonate case, plus a U.FL connector for an external active antenna (see the two antenna parts below) if the signal needs a boost. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314>

CHF 35.80

SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20 Pack)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 15 × 4 × 0.3 cm · 25 g



neu SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20 Pack) · PRT-12796 @ Mouser

SparkFun @ Mouser · in stock

Female-to-female DuPont jumper wires, 6 in (~150 mm), 20-pack. Used in the solderless GPS bring-up for: (a) 3× links between the 1.27→2.54 mm SWD adapter PCB's 2.54 mm side and the SparkFun MAX-M10S breakout's UART header (TX/RX/GND); (b) 1× link from the JP4 7-pin female header's 3.3 V pin to the GPS VCC pin. You need 4 jumpers from the pack; the rest stay in your bin for the next project. Also useful for any future I²C / Qwiic-jumper experiments on the LilyGO build. · SMA / U.FL · firmware: open-source

<https://www.mouser.ch/ProductDetail/SparkFun/PRT-12796?qs=WYAARYrbSnYazPSZHWXcfQ%3D%3D>

CHF 2.19

u-blox MAX-M10S GNSS module

OSS firmware

UART pins

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

☐ Motion sensors

☐ SD-card

L×B×H 1.01 × 0.97 × 0.25 cm · 0.60 g

Datasheet: https://content.u-blox.com/sites/default/files/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu u-blox MAX-M10S GNSS module · MAX-M10S-00B @ DigiKey

u-blox @ DigiKey · in stock

Wired to UART4 @ 38400. The firmware's GPS source. · UART pins · firmware: open-source

<https://www.digikey.com/en/products/detail/u-blox/MAX-M10S-00B/15712906>

CHF 10.38

Enclosure 7

Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 12 × 9 × 6 cm · 154 g



neu Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) · 1554G2GYCL @ DigiKey

Hammond Manufacturing @ DigiKey · in stock

Polycarbonate IP66 project box, external 120 × 90 × 60 mm, clear PC lid. Lid sealed with O-ring gasket and four corner screws. RF-transparent so onboard GPS (and any radar) reads through the wall. The Hall sensor reads a passing magnet through the lid, so you can power-cycle the recorder without opening the box. Leave the wall un-drilled to keep the seal intact and charge via Qi wireless through the wall (the STEVAL supports it). If the LPS22DF barometer needs ambient pressure, add a Gore-type vent. IP66 — splash and water-jet rated, NOT submersible. If the box is mounted on a foilboard that will wipeout-submerge, prefer the SERPAC RBF63 (IP67) below. · enclosure · firmware: open-source

<https://www.digikey.com/en/products/detail/hammond-manufacturing/1554G2GYCL/2359944>

CHF 22.77

SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

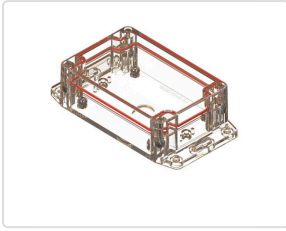
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 12 × 8 × 4 cm · 75 g



neu SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm) · RBF53P06C10C @ DigiKey

SERPAC @ DigiKey · in stock

Polycarbonate IP67 project box, external 120 × 80 × 40 mm (4.72 × 3.15 × 1.59 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. IP67 = submersion to 1 m for 30 min, the right call for a foilboard recorder that will go under during wipeouts. Smaller footprint than the RBF63 line — the MovementLogger (STEVAL 63 × 40 mm + SparkFun MAX-M10S breakout 25 × 25 mm side-by-side ≈ 88 × 40 mm) fits cleanly inside the ~108 × 65 mm interior with no excess case length to flop around. RF-transparent (GPS works through the wall), Hall-sensor magnet flips through the lid, Qi-charge through the wall — same tricks as the Hammond and the RBF63 line. · enclosure · firmware: open-source

<https://www.digikey.com/en/products/detail/serpac/RBF53P06C10C/17100309>

CHF 21.97

Solderless alternative — bring-up only

If you don't want to touch the iron yet (or you want to verify the wiring before committing to a permanent build), the BOM ships a single-SKU AliExpress turnkey cable that plugs into JP2 without soldering, plus a 2-part power-tap kit for JP4. **Caveat:** a vibrating pumpfoil board can wiggle the cable loose mid-session — use this for bring-up / development, switch to the soldered rig below for production.

Reference videos (watch first)

- [ControllersTech — GPS Module and STM32 \(NEO-6M\)](#): generic STM32 + u-blox GPS over UART. NEO-6M but the wiring is identical to MAX-M10S (TX↔RX crossover, GND, 3.3 V). Watch this one if you watch one.
- [SparkFun — MAX-M10S Product Showcase](#): the specific GPS breakout from this BOM (PRT-18037). Shows the U.FL connector and the UART header pins you'll connect to.
- [Chris Park — STM32 + u-blox UART NMEA, wiring from 11:33](#): the closest-fit hookup demo. Uses M8N (same pinout as the MAX-M10S), shows 4 DuPont female leads plugged onto VCC/GND/TX/RX on the GPS module. Translate directly to the STEVAL: JP4 → 3 V & GND, JP2 → TX/RX (crossover: STEVAL TX → GPS RX, STEVAL RX → GPS TX).

There is no exact "STEVAL-MKBOXPRO + solderless JTAG cable + MAX-M10S" video — that combination is bespoke to this build. The two videos above cover the load-bearing concepts (UART wiring + the specific GPS module).

Shopping list (3 SKUs, ~CHF 19 total)

#	Part	Where / Price
1	AliExpress FC 14-pin Cable — 1.27 mm IDC → DuPont female, 30 cm, select 2X7P variant (<i>BOM card above</i>). Handles the JP2 ↔ GPS data path end-to-end (TX, RX, GND on 3 of 14 wires; the others stay unused).	AliExpress item 1005011846708876 · CHF 9.69 + CHF 4.65 shipping = ~CHF 14 · 2–3 weeks to CH
2	Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (<i>BOM card above</i>). Slips onto JP4 to expose the 3.3 V tap for GPS VCC.	DigiKey CH · ~CHF 1
3	SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20-pack; you only need 1 for the JP4 → GPS VCC link) (<i>BOM card above</i>). The other 19 stay in your maker bin.	DigiKey CH · ~CHF 4

Earlier BOM iteration carried an all-DigiKey 4-SKU path (Samtec FFSD-07 + 1.27→2.54 adapter PCB + DuPont jumpers + this header) for buyers who refuse to wait for AliExpress shipping. The Samtec FFSD-07 was dropped once the AliExpress listing was browser-verified as stable with a real product photo and a 14-pin 2X7P variant — the AliExpress cable is now the canonical solderless data path. If the AliExpress listing ever dies, the all-DigiKey assembly recipe is in the git log (commit ada0363 + parent for the Samtec note).

Step 1 · JP4 power tap (3.3 V)

1. The JP4 connector is the 7-pin 2.54 mm strip on the side of the board (Multicomp MC-SVT1-S07-G).
2. **Set the JP4 domain switch to 3 V** (not 1.8 V).
3. Power on the box and **meter the JP4 3 V pin**: must read **3.25–3.40 V**. Power off again.
4. Slip a standard **2.54 mm 7-pin female header strip** onto JP4 (no solder — slides on). DuPont jumper from the 3 V pin → GPS VCC.

Step 2 · JP2 data lines (UART4)

1. Plug the JTAG cable into JP2 (FTSH-107 keyed/shrouded socket — only goes on one way).
2. Wire 3 DuPont jumpers from the adapter PCB's 2.54 mm side to the **SparkFun MAX-M10S breakout's UART header**:

STEVAL JP2 pin	Net (MCU)	→	SparkFun GPS pin
13	UART4_RX / PA1	→	TX
14	UART4_TX / PA0	→	RX
7 or 11	GND	→	GND

Note the TX/RX crossover (each end's "TX" pairs with the other's "RX"). The SparkFun breakout's UART header has male 2.54 mm pins populated — female DuPont leads slide on directly. **Do not touch SWD pins (4 / 6 / 8 / 10 / 12)** — they're the debug bus; bridging one means bricked debug access.

Step 3 · Bring-up test

1. Insert the SD card. Take the box **outdoors, clear sky**. Power on.
2. Cold fix is typically reached in **~30–60 s**. Confirm with a growing `GpsNNN.csv` on the SD card, or the BLE `SensorStream gps_valid` flag, or the antenna-test buzzer build.
3. No fix in a few minutes? **Swap pin 13 ↔ pin 14** — reversed TX/RX is by far the most common wiring mistake and swapping is harmless. Then re-check GND continuity and GPS VCC = 3.3 V under load.

When to switch to soldering: once bring-up works and you're ready for water trials. Vibration + spray + a loose cable will end a session early; the soldered rig below is the durable answer.

Soldering the u-blox MAX-M10S GPS to the STEVAL-MKBOXPRO

The MovementLogger firmware reads GPS over **UART4** on MCU pins **PA0 (TX)** / **PA1 (RX)** at **38400 baud, 8N1, 3.3 V logic**. The MAX-M10S is **not** on the box — it must be wired by hand to **JP2**, the 14-pin programming connector (Samtec FTSH-107-01-L-D, 1.27 mm pitch). Full source: [GPS SOLDERING.md](#).

Tools you need

- Fine-conical-tip iron, flux, magnification (10× loupe or microscope), tweezers, steady bench.
- **30 AWG** enamelled / wire-wrap wire (≤ 0.5 mm), four lengths.
- Multimeter (continuity + DC volts).
- Kapton tape + a dab of hot glue for strain relief.

JP2 pinout — pin 1 = square pad / silk triangle

Pin	Net	Use?
1	1V8 rail (JTAG ref)	NOT power — 1.8 V, weak
4 / 6 / 8 / 10 / 12	SWDIO / SWCLK / SWO / JTDI / NRST	Leave alone (debug bus)
7	GND	OK for GPS GND
11	GND	OK for GPS GND
13	UART4_RX (PA1)	GPS TX → here
14	UART4_TX (PA0)	GPS RX → here

Pins 3, 5, 9 carry no signal — leave unconnected. **Meter-verify** pin 1 against the silkscreen triangle and continuity-check every pin before soldering — connector orientation differs between board builds.

Wiring — note the TX/RX crossover

GPS pin	→	JP2 pin	Net (MCU)
TX (GPS out)	→	13	UART4_RX (PA1)
RX (GPS in)	→	14	UART4_TX (PA0)
GND	→	7 or 11 (meter-verified)	GND
VCC 3.3 V	→	JP4 3 V pin (not JP2!)	Board 3 V rail

GPS power — 3.3 V from JP4

- JP4 is the 7-pin extension-board connector (Multicomp MC-SVT1-S07-G). Set its domain switch to **3 V** (not 1.8 V).
- Power the box and **meter the JP4 3 V pin** — must read **3.25–3.40 V** before connecting GPS VCC. Power down again before soldering VCC.
- GPS GND and box GND must be common — take GND from JP4 or a JP2 GND pin.
- No graceful shutdown: the Hall-sensor magnet cuts the whole supply rail at once (firmware F-PWR-5). The on-SD format tolerates it.

Safety — read before the iron is hot

- **3.3 V logic only.** PA0/PA1 are **not 5 V tolerant**. A 5 V GPS breakout's TX into PA1 will destroy the MCU — level-shift or use a 3.3 V-native MAX-M10S carrier (e.g. the SparkFun breakout above).
- **Never bridge to the SWD pins** (4/6/8/10/12) — solder bridge → bricked debug access.
- Keep each pad heated < 2 s. FTSH-107 pads lift if cooked.
- **Strain-relieve all four wires** with Kapton + hot glue close to JP2 — a tugged wire rips the pad off the PCB.

Procedure

1. **Power down completely** — switch S1 off, USB unplugged, LiPo disconnected.
2. **Locate JP2** (14-pin fine-pitch header by the MCU). Identify pin 1.
3. **Continuity-check before soldering** (meter, board off): JP2 pin 13 ↔ MCU PA1, JP2 pin 14 ↔ MCU PA0, candidate GND pin ↔ battery (–). Confirm, don't assume.
4. **Pre-tin** the three JP2 pads and the wire ends with flux.
5. **Solder the data wires:** pin 13 ← GPS TX, pin 14 ← GPS RX. Then **GND**.
6. **GPS VCC last:** with the board powered, confirm 3.3 V at the JP4 3 V pin, power down again, solder the VCC wire.
7. **Inspect under magnification** for bridges — especially pin 14 to pin 12 and onto the SWD pins.
8. **Strain-relieve** all four wires (Kapton + hot glue).

Bring-up test

1. Insert the SD card, take the box **outdoors with clear sky**, power on.
2. The firmware sends the u-blox UBX config on boot, then expects 10 Hz NMEA. First fix is typically reached within **~30–60 s cold**.
3. Confirm a fix by **any** of: a growing GpsNNN.csv on the SD card · the BLE SensorStream gps_valid flag going set · (antenna-test build, make GPS_ANTENNA_BEEP=1) the buzzer beeping N times every 10 s with N = satellites.

No fix after several minutes outdoors? — in order of likelihood

1. **Swap the two data wires** (pin 13 ↔ pin 14). TX/RX reversed is the single most common wiring mistake and swapping is harmless to try.
2. Re-check **GND** continuity (cold/again) and that GPS VCC really is 3.3 V *at the module* under load.
3. Confirm the module is a 3.3 V part and actually powered (most MAX-M10S carriers have a power LED).
4. Give it a longer cold-start with an unobstructed sky view; indoors it will never fix.